

Electrical and Electronics Measuring Instruments	L	P	C
	4		4

Discipline(s) / EAE / OAE	Semester	Group	Sub-group	Paper Code
EE/EEE	5	PC	PC	EEC-305

Marking Scheme:												
1. Teachers Continuous Evaluation: 25 marks												
2. Term end Theory Examinations: 75 marks												
Instructions for paper setter:												
1. There should be 9 questions in the term end examinations question paper.												
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.												
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.												
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.												
5. The requirement of (scientific) calculators / log-tables / data – tables may be specified if required.												
Course Objectives :												
1.	Introduce how to measure power and energy											
2.	Understand working and applications of potentiometers and bridges											
3.	Knowledge to use printers and recorders											
4.	Selection of proper type and specification of transducers											
Course Outcomes (CO)												
CO 1	Identify and classify various types of instruments for power and energy measurement											
CO 2	Develop the knowledge of working and applications of potentiometers and bridges											
CO 3	Ability to apply proper recorder and printer in measurement											
CO 4	Describe working principle selection criteria and application of various transducers in measurement system.											
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)												
	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO 1	2	3	2	3	3	2	2	1	1	1	1	3
CO 2	3	3	2	3	3	2	2	1	1	1	1	3
CO 3	3	3	3	3	3	2	2	1	1	1	1	3
CO 4	3	3	3	3	3	2	2	1	1	1	1	3
UNIT I												
Power and Energy Measurement												
Instrument transformers: , CT and PT, Ratio and phase angle errors.												
Measurement of Power: Single phase and three phase dynamometer wattmeter, LPF and UPF, expression for deflecting and control torques. Type of P.F. Meters, dynamometer and moving iron type, Frequency meters, Resonance type and Weston type, synchroscopes.												
Measurement of Energy: single phase and three phase induction type energy meter, driving and braking torques, errors and compensations, testing by phantom loading, trivector meter, maximum demand meters.												
UNIT II												
Potentiometers and Bridges: Basics of DC potentiometers, laboratory type potentiometer, multi range potentiometer, applications of DC potentiometers, AC potentiometer; polar and coordinate type,, Drysdale polar potentiometer. Gall Trinsley potentiometer. applications of AC potentiometer. Bridges for measuring												

low, medium and high resistance; Carey Foster's bridge, Kelvin's double bridge, Mega ohm bridge, Megger.

A.C. Bridges: Measurement of inductance and capacitance, Maxwell's bridge, Hay's bridge, Anderson's bridge, Owen's bridge, Heaviside Bridge and its modifications, Desauty bridge. Wien's bridge, Schering Bridge..

UNIT III

Display Devices and Recorders: Introduction of various display devices, LCD, LED and plasma display, CRO & its applications, measurement of frequency (lissajous patterns), sampling oscilloscope, DSO. Recorders: requirement of recording data, selection of recorder for a particular application, analog, graphic, strip chart, galvanometeric, circular chart, XY, digital recorders, single point and multipoint recorders.

Printers: Types of Printers, Drum type printer, dot matrix type printer, Ink-jet and Laser jet printers

UNIT IV

Transducers: Introduction and Classification of Transducers. Primary and secondary sensing elements, Basic Working principle and applications of resistive, inductive and capacitive transducers. Working principle and applications of. LVDT, RTD, Thermistor, piezoresistors, strain gauze, angular velocity transducers, opto electronic transducers, inverse transducers

Textbook(s):

1. D. Patranabis, "Sensors and Transducers", PHI Learning Pvt. Ltd., 2nd edition
2. D V S Murty, "Transducers and Instrumentation", PHI Learning Pvt. Ltd.
3. E. W. Golding and F. C. Widdis - Electrical Measurements and measuring Instruments, Wheeler Publishing, 5th Ed..
4. A. K. Sahwney - Electrical & Electronic Measurement & Instrumentation, Dhanpat Rai & Co., 2000

Reference Books:

1. Buckingham and Price - Principles of Electrical Measurements, Prentice Hall, 1970
2. Reissland, M. U. - Electrical Measurements: Fundamentals, Concepts, Applications New Age.
3. W. D. Cooper, "Modern Electronic Instrumentation & Measurement Technique" PHI, 1998